Waste Management 38 (2015) 102–104

Contents lists available at ScienceDirect

Waste Management

journal homepage: www.elsevier.com/locate/wasman

Quick-start of full-scale anaerobic digestion (AD) using aeration

Anders Lagerkvist *, Markku Pelkonen, Tommy Wikström

Waste Science and Technology, Luleå University of Technology, 97187 Luleå, Sweden

ARTICLE INFO

Article history:
Received 8 July 2014
Accepted 16 December 2014
Available online 27 January 2015

Keywords:
Start-up
Anaerobic digestion
Sewage sludge
Food-waste
Methanogens

ABSTRACT

A conventional 1300 m^3 continuously stirred anaerobic tank reactor at the city of Boden, north Sweden, which was receiving a feed of both sewage sludge and food waste, was put out of operation due to the build-up of a float phase. The reactor was emptied and cleaned. At start-up there was no methanogenic sludge available, so an unconventional start-up procedure was applied: The reactor was rapidly (8 days with 1200 kg of total solids (TS) added daily) filled with thickened, and slightly acidic sewage sludge, showing only slight methane generation, which was subsequently heated to 55 °C. Then compressed air was blown into the digester and within a month a fully functional methanogenic culture was established. The transfer from acidogenic to methanogenic conditions happened in about one week. As a start-up technique this is fast and cost efficient, it only requires the access of a compressor, electricity and a source of air. In total, about 16 tonnes of oxygen were used. It is proposed that this method may also be used as an operational amendment technique, should a reactor tend to acidify.

© 2015 Elsevier Ltd. All rights reserved.

1. Introduction

At present anaerobic treatment of wastes is going through a revival in Sweden as it has been identified as a significant and renewable source of vehicle fuel. A similar wave of interest was also apparent in the 1980's, following a hike in oil prices in 1973 and 1979, but today the emphasis is more clearly on the reduction of fossil carbon emissions and its associated impact on climate. The number of new plants and digestion capacity is increasing, which increases the demand of substrate beyond the easily accessible and easily treatable ones. One likely impact of the present trends is that operational problems will increase, especially when using the traditional continuously stirred tank reactor developed for sewage sludge digestion, and also the need of reliable and expedient start-up procedures will also be increasing. Well established start-up measures for Anaerobic Digestion (AD) includes seeding (Fernández et al., 2001; Syutsubo et al., 1997), dilution of substrates, keeping the temperature stable, gradually increasing the load and waiting. Start-up times vary a lot but can stretch over months. A common problem is acid over-load. In recent research only a few papers focusing on start-up procedures has been published, e.g. Zeeman et al. (1988) although it is touched upon in other studies, e.g. in Ahring (1994).

In landfill research, it was found in several studies during the 1980's and 90's that methanogenic conditions may be created, in a large variation of substrates, by aerobic pre-treatment. Aeration has been applied in various ways, e.g. through the use of composted waste as bottom liners in landfill cells or forced aeration. Stimulating the onset of methane formation was a theme because the need to avoid acid landfill leachate, and its associated transport of hazardous trace substances, was well recognised. Different aspects of aerobic pre-treatment were discussed e.g. by Damiecki (1987), Lagerkvist and Chen (1993) and Spendlin (1991); and it has been used since in solid waste treatment. A recent example is Charles et al. (2009). However, this knowledge does not seem to have resulted in any application in the CSTR-community, which could be observed in published literature.

Given the opportunity through a full-scale reactor break-down and the absence of a suitable inoculum, a full-scale test of using forced aeration as a start-up enhancement was done. The results including a microbiological characterization is discussed here.

The digester at Swedjans sewage treatment plant by Boden, Sweden, is a CSTR of 1300 m^3 operated at about 55 °C. It was originally designed to digest sewage sludge only, but has over the last decade been fitted with an additional feed line for various organic wastes, primarily source-separated kitchen waste. In the fall of 2011, it had accumulated too much irrelevant material (Fig. 1) from the wastes that it ceased to function.

2. Materials and methods

2.1. Inoculum

Due to a long standstill, there was no methanogenic inoculum left for start-up, only about 1200 kg of TS per day of thickened

* Corresponding author. Tel.: +46 920491908.
E-mail address: al@ltu.se (A. Lagerkvist).

http://dx.doi.org/10.1016/j.wasman.2014.12.016
0956-053X/© 2015 Elsevier Ltd. All rights reserved.



Fig. 1. Residual material accumulated in digester and causing malfunction. Its main constituents are fibrous materials and plastics.

and slightly acidic, pH about 5.5, sewage sludge. The sludge was a mixture from pre-settling and post aeration settling basins, about equal volumes of each. The total solids (TS) content of the sludge in regular operation is about 4%, but in order to fill the reactor quicker a less thickened sludge was used. At the start of aeration the digester content had a TS of about 2%, and the VS was 70% of TS. At day zero, the digester had been filled with thickened sludge that had been heated to 55 °C.

2.2. Experimental procedures

After the desired operation temperature of 55 °C had been reached, an air compressor was connected to the circulation loop for sludge, at the digester bottom. Initially, 3.33 l per second of compressed air (7 bar) was added for one week and then interrupted four days due to equipment malfunction. After that the rate of gas addition was kept at this rate and at day 14 it was increased to 10 l per second on working hours and 5 l per second during the rest of the days, i e 6.67 l per second on average over a 24 h period. From day 20 to day 22, the aeration was held constant at 6.67 l per second until it was stopped at day 23. The total air addition was about 50,000 N m^3 or 16 tonnes of oxygen (at 20 °C).

2.3. Sampling and analyses

During the whole experiment data from the recirculation of the digester sludge was taken for TS, VS, pH and FISH-samples. Sam-

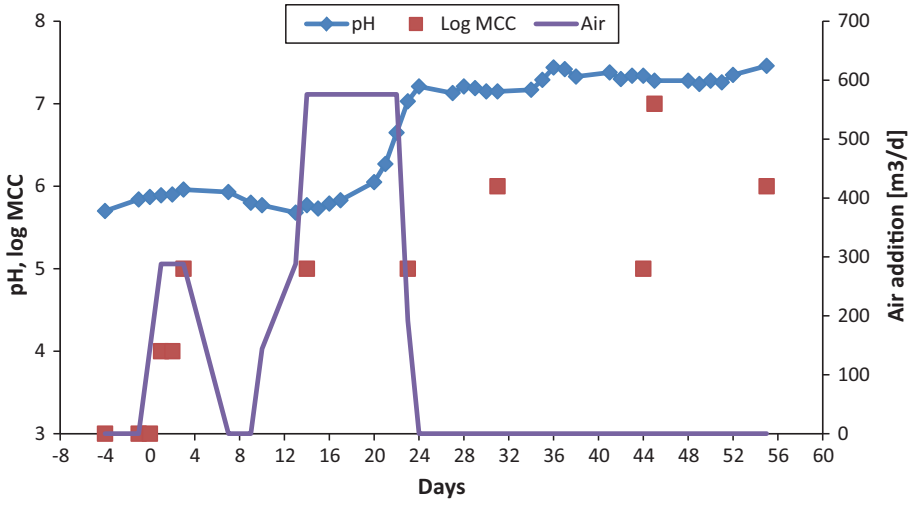


Fig. 2. Air addition and the development of pH and methanogens during start-up. The initial MCC values of 10^3 are "less then" values.

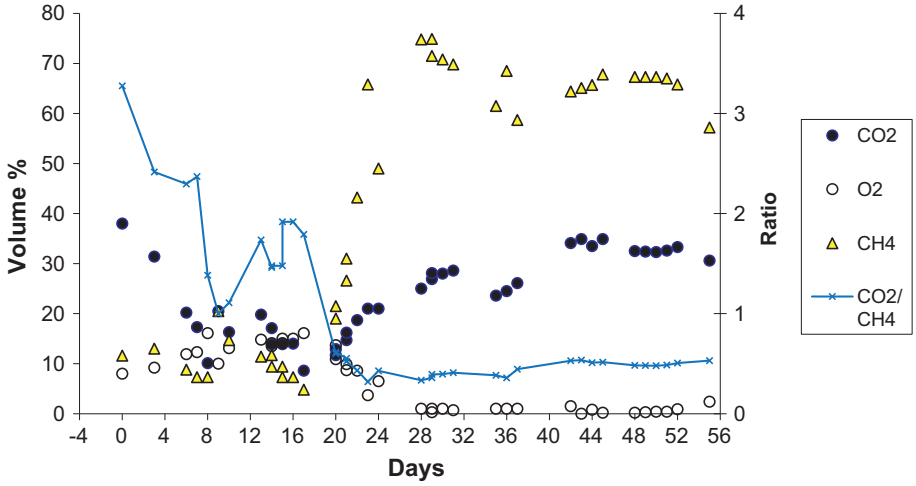


Fig. 3. Gas composition at the digester gas outlet.

ples were also taken directly from the gas outlet of the digester as a complement to the online measurements.

The aeration was maintained by a compressor operating at 7 bar and using indoor air that was entered without dispersion at the bottom of the digester.

Total and volatile solids were analyzed according to SIS (1981).

The methanogen cell count (MCC) was established by the following routine: A 250 ml sample was homogenized using a Blentec Xpress model ES3 blender for $2 * 30$ s. Diluted samples were then fixed with paraformaldehyde and stored at −18 C. Fluorescent in-situ hybridization was performed as described in Blackall (1999). A general probe for Archaea, ARC915, was used to identify all methanogens (Crocetti, 2006). The probes were labeled at 5′-end with an indocyanine dye (Cy3). Microscope used was Diaphot-300 epifluorescence microscope (Nikon, Japan) with 100 W mercury lamp, equipped with filter sets (Chroma Tech, USA), and used with 1000× magnification. 10–15 fields of view were counted for each sample.

Gas samples were analysed *in situ* using an infrared gas analyzer (GEM 2000 Plus, Geotech) and in the lab using a Clarus 580 GC (PerkinElmer).

3. Results and discussion

The results are presented in Figs. 2 and 3. The development of pH, gas composition and methanogens were used as independent response variables to observe the impact of aeration and the trends of all three variables coincide.

The air addition was stopped as the pH of the digester reached 7, and the feed of substrate was started one week after that. The variable, and fairly low MCC in that phase is probably due to the digester feed additions, which were somewhat varying and on the average of low TS-content. However, the difference between pre- and post aeration are clear. The responses of the pH and gas composition to the unintended interruption of aeration around day 8 indicate the reversibility of the process in its early stages.

The aerobic degradation of organic acids leads to the elevation of the pH, thus making already present methanogens able to obtain energy and grow. The degradation is also indicated by total TS, and VS-observations, but less clear. E.g. the observed VS at day 7 was about 14.5 g per l, and at day 21 it was 12 g per l. The VS/TS-ratio for the same days were 0.72 and 0.7. However, in spite of clearly sinking trends over time, the TS and VS data were varying day to day, most likely due to the small (250 ml) samples used.

It can be noted that the methane concentration of the collected gas was already about 10% as the aeration was started. So should the development of a methanogenic flora have progressed regardless of the aeration? Is it just a coincidence that aeration was done at the same time as the methanogens thrived? Experience indicates that it is not (Lagerkvist, 1994) and that there is a dynamic balance between acidification and the development of a predominantly methanogenic degradation. Also, the trends of pH and gas composition during the interruption of aeration indicates that a return towards acidogenic conditions were on-going. However, the occurrence of methane in a basically acidogenic environment indicates the heterogeneity that is typical for a three phase system, there are most often micro-environments available for different processes.

4. Conclusions

Aeration showed to be an efficient method of establishing methanogenic conditions in a full- scale AD-reactor. Compared to the common technique of slow adaption, it is much more expedient. This technique was applied in landfill research and practice already in the 1980's, and it has now also been shown in a full-scale CSTR AD system. It is probable that the technique can be applied to any three phase system (particles, liquid and gas) as long as there are microenvironments where methanogens can grow uninhibited by the oxygen content of the macro environment. This start-up technique is cost efficient compared to alternative methods, and it could probably also be used for amending occasional acidification problems during regular operation of digesters.

Acknowledgements

The co-operation from the staff of Svedjans reningsverk at the city of Boden is gratefully acknowledged as well as the financial support from EU target 2 structural funds, the city of Boden, the Norrbottens läns landsting, Norrbottens länsstyrelse and LTU.

References

Ahring, B.K., 1994. Status on science and application of thermophilic anaerobic digestion. Water Sci. Technol. 30 (12), 241–249.

Blackall, L., 1999. Practical methods for the examination and characterization of activated sludge. In: Seviour, Robert J., Blackall, L. (Eds.), The Microbiology of Activated Sludge. Springer.

Charles, W., Walker, L., Cord-Ruwisch, R., 2009. Effect of pre-aeration and inoculum on the start-up of batch thermophilic anaerobic digestion of municipal solid waste. Bioresour. Technol. 100 (8), 2329–2335.

Crocetti, G., 2006. An update and optimisation of oligonucleotide probes targeting methanogenic archaea for use in fluorescence in situ hybridization (FISH). J. Microbiol. Methods 65 (1), 194–201.

Damiecki, R., 1987. Vorbehandlung des Sickerwassers innerhalb der Deponie. ENTSORGA-Magazin 5, 41–44.

Fernández, B., Porrier, P., Chamy, R., 2001. Effect of inoculums-substrate ratio on the start-up of solid waste anaerobic digesters. Water Sci. Technol. 44 (4), 103–108.

Lagerkvist, A., Chen, H., 1993. Control of two-step anaerobic degradation of municipal solid waste (MSW) by enzyme addition. Water Sci. Technol. 27 (2), 47–56.

Lagerkvist, A., 1994. Experiences of two-step anaerobic degradation of MSW in laboratory and field experiments. In: Seventeenth International Madison Waste Conference, 21–22/9, Madison, Wisconsin, pp. 241–259.

SIS, 1981. SS 028113 determination of dry matter and ignition residue in water, sludge and sediment. SIS Förlag, AB.

Spendlin, H.-H., 1991. Untersuchungen zur frühzeitigen Initiierung der Methanbildung bei festen Abfallstoffen. D Sc thesis, Economica Verlag, Bonn.

Syutsubo, K., Harada, H., Obashi, A., Suzuki, H., 1997. An effective start-up of thermophilic UASB reactor by seeding mesophilically-grown granular sludge. Water Sci. Technol. 36 (6–7), 391–398.

Zeeman, G., Sutter, K., Vens, T., Koster, M., Wellinger, A., 1988. Psychrophilic digestion of dairy cattle and pig manure: start-up procedures of batch, fed-batch and CSTR-type digesters. Biol. Wastes 26 (1988), 15–31.